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#include <iostream>
#include <iomanip>
#include <vector>
#include <algorithm>
using namespace std;

double balance = 0;

struct Transaction {
    int id;
    string type;
    double amount;
    string description;
};

vector<Transaction> transactions;
int nextId = 1;

string header() {
    string headerText = "********** Ethio Telecom **********\n\n";
    cout << headerText;
    return headerText;
}

void displayTransactions() {
    if (transactions.empty()) {
        cout << "No transactions found.\n";
        return;
    }
    cout << "\n=== Transaction History ===\n";
    for (const auto& t : transactions) {
        cout << "ID: " << t.id
             << " | Type: " << t.type
             << " | Amount: " << fixed << setprecision(2) << t.amount
             << " ETB | Desc: " << t.description << endl;
    }
}

void searchTransaction() {
    int id;
    cout << "Enter Transaction ID to search: ";
    cin >> id;
    auto it = find_if(transactions.begin(), transactions.end(),
                      [id](const Transaction& t) { return t.id == id; });
    if (it != transactions.end()) {
        cout << "\nTransaction Found:\n";
        cout << "ID: " << it->id
             << " | Type: " << it->type
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            << " | Amount: " << fixed << setprecision(2) << it->amount
            << " ETB | Desc: " << it->description << endl;
    } else {
        cout << "Transaction not found.\n";
    }
}

void updateTransaction() {
    int id;
    cout << "Enter Transaction ID to update: ";
    cin >> id;
    auto it = find_if(transactions.begin(), transactions.end(),
                      [id](const Transaction& t) { return t.id == id; });
    if (it != transactions.end()) {
        cout << "Enter new description: ";
        cin.ignore();
        getline(cin, it->description);
        cout << "Transaction updated successfully.\n";
    } else {
        cout << "Transaction not found.\n";
    }
}

void deleteTransaction() {
    int id;
    cout << "Enter Transaction ID to delete: ";
    cin >> id;
    auto it = remove_if(transactions.begin(), transactions.end(),
                        [id](const Transaction& t) { return t.id == id; });
    if (it != transactions.end()) {
        transactions.erase(it, transactions.end());
        cout << "Transaction deleted successfully.\n";
    } else {
        cout << "Transaction not found.\n";
    }
}

double buyPackage(double pac_price, const string& desc) {
    if (pac_price > balance) {
        cout << header() << "Insufficient balance\n";
    } else {
        balance -= pac_price;
        transactions.push_back({nextId++, "Purchase", pac_price, desc});
        cout << "Package purchased successfully. Your balance is "
            << fixed << setprecision(2) << balance << " ETB\n";
    }
    return balance;
}
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double recharge(int amount) {
    if (amount <= 0) {
        cout << header() << "Enter a valid amount\n";
    } else {
        balance += amount;
        transactions.push_back({nextId++, "Recharge", (double)amount, "Account
top-up"});
        cout << "Account recharged successfully. Your balance is "
             << fixed << setprecision(2) << balance << " ETB\n";
    }
    return balance;
}

void displayMainMenu() {
    header();
    cout << "1. Check Balance\n";
    cout << "2. Buy Package\n";
    cout << "3. Recharge Balance\n";
    cout << "4. View Transactions\n";
    cout << "5. Search Transaction\n";
    cout << "6. Update Transaction\n";
    cout << "7. Delete Transaction\n";
    cout << "8. Exit\n";
    cout << "Enter your choice: ";
}

void packageMenu() {
    int choice;
    do {
        header();
        cout << "1. Internet Package\n";
        cout << "2. Voice Package\n";
        cout << "3. SMS Package\n";
        cout << "4. Back to Main Menu\n";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1) {
            header();
            cout << "1. 1GB [15 ETB]\n";
            cout << "2. 5GB [50 ETB]\n";
            cout << "3. 10GB [75 ETB]\n";
            cout << "4. Back\n";
            cout << "Enter choice: ";
            cin >> choice;
            switch (choice) {
                case 1: buyPackage(15, "1GB Internet"); break;
                case 2: buyPackage(50, "5GB Internet"); break;
                case 3: buyPackage(75, "10GB Internet"); break;
                case 4: break;
                default: cout << "Invalid input\n";
            }
        } else if (choice == 2) {
            header();
            cout << "1. 1 hour [35 ETB]\n";
            cout << "2. 2 hours [50 ETB]\n";
            cout << "3. 5 hours [100 ETB]\n";
            cout << "4. Back\n";
            cout << "Enter choice: ";
            cin >> choice;
            switch (choice) {
                case 1: buyPackage(35, "1 Hour Voice"); break;
                case 2: buyPackage(50, "2 Hours Voice"); break;
                case 3: buyPackage(100, "5 Hours Voice"); break;
                case 4: break;
                default: cout << "Invalid input\n";
            }
        } else if (choice == 3) {
            header();
            cout << "1. 550 SMS [15 ETB]\n";
            cout << "2. 1500 SMS [30 ETB]\n";
            cout << "3. Unlimited SMS [75 ETB]\n";
            cout << "4. Back\n";
            cout << "Enter choice: ";
            cin >> choice;
            switch (choice) {
                case 1: buyPackage(15, "550 SMS"); break;
                case 2: buyPackage(30, "1500 SMS"); break;
                case 3: buyPackage(75, "Unlimited SMS"); break;
                case 4: break;
                default: cout << "Invalid input\n";
            }
        } else if (choice == 4) {
            break;
        } else {
            cout << "Invalid input\n";
        }
    } while (true);
}

int main() {
    cout <<
"@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@@@@@@@@@@@\n";
    cout << "@@——MIZAN TEPI UNIVERSITY————@@\n";
    cout << "@@—————TEPI CAMPUS—————@@\n";
    cout << "@@—SCHOOL OF COMPUTING AND INFORMATICS——@@\n";
    cout << "@@—DEPARTMENT OF INFORMATION TECHNOLOGY—@@\n";
    cout << "@@———GROUP SIX C++ PROJECT————@@\n";
    cout << "@@ Name                    ID           @@\n";
    cout << "@@1. Estifanos Getawey      8057/17      @@\n";
    cout << "@@2. Getachew Birhan       8018/17      @@\n";
    cout << "@@3. Abreham Tadess        7962/17      @@\n";
    cout << "@@4. Abule Dulo           8968/17      @@\n";
    cout << "@@5. Asaye Birhan          8279/17      @@\n";
    cout << "@@                                  @@\n";
    cout << "@@       SUBMITTED TO: MR. ABRIHAM         @@\n";
    cout << "@@       SUBMISSION DATE: [DATE]           @@\n";
    cout <<
"@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@@@@@@@@@@@\n\n";

    int choice;
    do {
        displayMainMenu();
        cin >> choice;

        switch (choice) {
            case 1:
                header();
                cout << "Your balance: " << fixed << setprecision(2) << balance << "
ETB\n";
                break;
            case 2:
                packageMenu();
                break;
            case 3:
                int amount;
                header();
                cout << "Enter amount to recharge: ";
                cin >> amount;
                recharge(amount);
                break;
            case 4:
                displayTransactions();
                break;
            case 5:
                searchTransaction();
                break;
            case 6:
                updateTransaction();
                break;
            case 7:
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            deleteTransaction();
            break;
        case 8:
            cout << "Exiting... Thank you!\n";
            break;
        default:
            cout << "Invalid choice. Try again.\n";
    }
    cout << endl;
} while (choice != 8);

return 0;
}
```